In [7]:
```python
import pandas as pd
```

In [8]:
```python
df = pd.read_csv("data_science_job.csv")
```

In [9]:
```python
df
```

Out[9]:

	enrollee_id	city	city_development_index	gender	relevent_experience	enrol
0	8949	city_103	0.920	Male	Has relevent experience	
1	29725	city_40	0.776	Male	No relevent experience	
2	11561	city_21	0.624	NaN	No relevent experience	F
3	33241	city_115	0.789	NaN	No relevent experience	
4	666	city_162	0.767	Male	Has relevent experience	
...	...	...	...	...	...	
19153	7386	city_173	0.878	Male	No relevent experience	
19154	31398	city_103	0.920	Male	Has relevent experience	
19155	24576	city_103	0.920	Male	Has relevent experience	
19156	5756	city_65	0.802	Male	Has relevent experience	
19157	23834	city_67	0.855	NaN	No relevent experience	

19158 rows × 13 columns

In [7]:
```python
df.shape
```

Out[7]:
```
(19158, 13)
```

In [8]:
```python
df.head
```

Out[8]:

```
<bound method NDFrame.head of        enrollee_id      city  city_development_in
dex gender  \
0             8949  city_103                     0.920   Male
1            29725   city_40                     0.776   Male
2            11561   city_21                     0.624    NaN
3            33241  city_115                     0.789    NaN
4              666  city_162                     0.767   Male
...            ...       ...                       ...    ...
19153         7386  city_173                     0.878   Male
19154        31398  city_103                     0.920   Male
19155        24576  city_103                     0.920   Male
19156         5756   city_65                     0.802   Male
19157        23834   city_67                     0.855    NaN

           relevent_experience enrolled_university education_level  \
0      Has relevent experience       no_enrollment        Graduate
1       No relevent experience       no_enrollment        Graduate
2       No relevent experience    Full time course        Graduate
3       No relevent experience                 NaN        Graduate
4      Has relevent experience       no_enrollment         Masters
...                        ...                 ...             ...
19153   No relevent experience       no_enrollment        Graduate
19154  Has relevent experience       no_enrollment        Graduate
19155  Has relevent experience       no_enrollment        Graduate
19156  Has relevent experience       no_enrollment     High School
19157   No relevent experience       no_enrollment  Primary School

      major_discipline  experience company_size    company_type  \
0                 STEM        20.0          NaN             NaN
1                 STEM        15.0        50-99         Pvt Ltd
2                 STEM         5.0          NaN             NaN
3      Business Degree         0.0          NaN         Pvt Ltd
4                 STEM        20.0        50-99  Funded Startup
...                ...         ...          ...             ...
19153       Humanities        14.0          NaN             NaN
19154             STEM        14.0          NaN             NaN
19155             STEM        20.0        50-99         Pvt Ltd
19156              NaN         0.0      500-999         Pvt Ltd
19157              NaN         2.0          NaN             NaN

       training_hours  target
0                36.0     1.0
1                47.0     0.0
2                83.0     0.0
3                52.0     1.0
4                 8.0     0.0
...               ...     ...
19153            42.0     1.0
19154            52.0     1.0
19155            44.0     0.0
19156            97.0     0.0
19157           127.0     0.0

[19158 rows x 13 columns]>
```

In [9]:

```
df.tail()
```

Out[9]:

	enrollee_id	city	city_development_index	gender	relevent_experience	enrol
19153	7386	city_173	0.878	Male	No relevent experience	
19154	31398	city_103	0.920	Male	Has relevent experience	
19155	24576	city_103	0.920	Male	Has relevent experience	
19156	5756	city_65	0.802	Male	Has relevent experience	
19157	23834	city_67	0.855	NaN	No relevent experience	

In [10]:

```
df.describe()
```

Out[10]:

	enrollee_id	city_development_index	experience	training_hours	targe
count	19158.000000	18679.000000	19093.000000	18392.000000	19158.00000
mean	16875.358179	0.828951	9.928036	65.185787	0.24934
std	9616.292592	0.123334	6.505268	59.885626	0.43264
min	1.000000	0.448000	0.000000	1.000000	0.00000
25%	8554.250000	0.740000	4.000000	23.000000	0.00000
50%	16982.500000	0.903000	9.000000	47.000000	0.00000
75%	25169.750000	0.920000	16.000000	88.000000	0.00000
max	33380.000000	0.949000	20.000000	336.000000	1.00000

In [11]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19158 entries, 0 to 19157
Data columns (total 13 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   enrollee_id             19158 non-null  int64  
 1   city                    19158 non-null  object 
 2   city_development_index  18679 non-null  float64
 3   gender                  14650 non-null  object 
 4   relevent_experience     19158 non-null  object 
 5   enrolled_university     18772 non-null  object 
 6   education_level         18698 non-null  object 
 7   major_discipline        16345 non-null  object 
 8   experience              19093 non-null  float64
 9   company_size            13220 non-null  object 
 10  company_type            13018 non-null  object 
 11  training_hours          18392 non-null  float64
 12  target                  19158 non-null  float64
dtypes: float64(4), int64(1), object(8)
memory usage: 1.9+ MB
```

In [13]:
```
df.isnull()
```

Out[13]:

	enrollee_id	city	city_development_index	gender	relevent_experience	enrollec
0	False	False	False	False	False	
1	False	False	False	False	False	
2	False	False	False	True	False	
3	False	False	False	True	False	
4	False	False	False	False	False	
...	...	...	...	...	...	
19153	False	False	False	False	False	
19154	False	False	False	False	False	
19155	False	False	False	False	False	
19156	False	False	False	False	False	
19157	False	False	False	True	False	

19158 rows × 13 columns

In [15]:
```
df[100:200]
```

Out[15]:

	enrollee_id	city	city_development_index	gender	relevent_experience	enrolle
100	6172	city_11	0.550	Male	Has relevent experience	nc
101	14672	city_173	0.878	NaN	No relevent experience	nc
102	18257	city_103	0.920	Female	Has relevent experience	nc
103	24493	city_103	0.920	NaN	Has relevent experience	nc
104	1180	city_16	0.910	Male	Has relevent experience	nc
...	...	...	...	...	...	
195	9890	city_103	0.920	Male	Has relevent experience	nc
196	32683	city_160	0.920	Male	Has relevent experience	nc
197	27104	city_36	0.893	Male	Has relevent experience	nc
198	7731	city_65	0.802	Male	Has relevent experience	nc
199	31943	city_71	0.884	Male	Has relevent experience	nc

100 rows × 13 columns

In [10]:

```
df1 = df[df['gender'] == 'Male']
```

In [14]:

```
df1
```

Out[14]:

	enrollee_id	city	city_development_index	gender	relevent_experience	enrol
0	8949	city_103	0.920	Male	Has relevent experience	
1	29725	city_40	0.776	Male	No relevent experience	
4	666	city_162	0.767	Male	Has relevent experience	
6	28806	city_160	0.920	Male	Has relevent experience	
7	402	city_46	0.762	Male	Has relevent experience	
...	...	...	...	...	...	
19151	11385	city_149	0.689	Male	No relevent experience	F
19153	7386	city_173	0.878	Male	No relevent experience	
19154	31398	city_103	0.920	Male	Has relevent experience	
19155	24576	city_103	0.920	Male	Has relevent experience	
19156	5756	city_65	0.802	Male	Has relevent experience	

13221 rows × 13 columns

In [34]:

```
import pandas as pd
```

In [35]:

```
import numpy as np
```

In [36]:

```
df = pd.read_csv("data_science_job.csv")
```

In [37]:

```
df
```

Out[37]:

	enrollee_id	city	city_development_index	gender	relevent_experience	enrol
0	8949	city_103	0.920	Male	Has relevent experience	
1	29725	city_40	0.776	Male	No relevent experience	
2	11561	city_21	0.624	NaN	No relevent experience	F
3	33241	city_115	0.789	NaN	No relevent experience	
4	666	city_162	0.767	Male	Has relevent experience	
...	...	...	...	...	...	
19153	7386	city_173	0.878	Male	No relevent experience	
19154	31398	city_103	0.920	Male	Has relevent experience	
19155	24576	city_103	0.920	Male	Has relevent experience	
19156	5756	city_65	0.802	Male	Has relevent experience	
19157	23834	city_67	0.855	NaN	No relevent experience	

19158 rows × 13 columns

In [38]:

```
df1 = df[df['gender'] == 'Male']
```

In [39]:

```
df1
```

Out[39]:

	enrollee_id	city	city_development_index	gender	relevent_experience	enrol
0	8949	city_103	0.920	Male	Has relevent experience	
1	29725	city_40	0.776	Male	No relevent experience	
4	666	city_162	0.767	Male	Has relevent experience	
6	28806	city_160	0.920	Male	Has relevent experience	
7	402	city_46	0.762	Male	Has relevent experience	
...	...	...	...	...	...	
19151	11385	city_149	0.689	Male	No relevent experience	F
19153	7386	city_173	0.878	Male	No relevent experience	
19154	31398	city_103	0.920	Male	Has relevent experience	
19155	24576	city_103	0.920	Male	Has relevent experience	
19156	5756	city_65	0.802	Male	Has relevent experience	

13221 rows × 13 columns

In [40]:
```
df.isnull().sum()
```

Out[40]:
```
enrollee_id                  0
city                         0
city_development_index     479
gender                    4508
relevent_experience          0
enrolled_university        386
education_level            460
major_discipline          2813
experience                  65
company_size              5938
company_type              6140
training_hours             766
target                       0
dtype: int64
```

In [43]:
```
df2 = df.dropna(axis = 0)
```

In [46]:
```
df2.shape
```

Out[46]:
```
(8434, 13)
```

In [50]:
```
from sklearn.impute import SimpleImputer
```

In [53]:
```
imp_mean = SimpleImputer(missing_values=np.nan, strategy='mean')
```

In [54]:
```
imp_mean.fit([[7,2,3], [4,np.nan,6], [10, 5, 9]])
```

Out[54]:

```
▾ SimpleImputer
▶ Parameters
```

In [55]:
```
x = [[np.nan, 2, 3], [4,np.nan,6], [10, np.nan, 9]]
```

In [56]:
```
x
```

Out[56]:
```
[[nan, 2, 3], [4, nan, 6], [10, nan, 9]]
```

In [66]:
```
imp_mean.transform(x)
```

Out[66]:
```
array([[ 7. ,  2. ,  3. ],
       [ 4. ,  3.5,  6. ],
       [10. ,  3.5,  9. ]])
```

In [67]:
```
#Creating Dataset
```

In [58]:
```
country = ["India", "france", "Germany", "france", "Germany", "India", "Germany"
```

In [59]:
```
len(country)
```

Out[59]:
```
10
```

In [60]:
```
age = [38, 43, 38, 34, 54, 'nan', 34, 56, 43,57]
```

In [61]:
```
len(age)
```

Out[61]:
```
10
```

In [68]:
```
salary = [34000, 45000, 54000, 'nan', 54500, 40000, 34000, 23000, 60000,23000]
```

In [69]:
```
len(salary)
```

Out[69]:
```
10
```

In [70]:
```
df_data = pd.DataFrame({'Country' : country, 'Age' : age, 'Salary' : salary})
```

In [71]:
```
df_data
```

Out[71]:

	Country	Age	Salary
0	India	38	34000
1	france	43	45000
2	Germany	38	54000
3	france	34	nan
4	Germany	54	54500
5	India	nan	40000
6	Germany	34	34000
7	france	56	23000
8	India	43	60000
9	Germany	57	23000

In [72]:
```
df_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 3 columns):
 #   Column   Non-Null Count  Dtype 
---  ------   --------------  ----- 
 0   Country  10 non-null     object
 1   Age      10 non-null     object
 2   Salary   10 non-null     object
dtypes: object(3)
memory usage: 372.0+ bytes
```

In [73]:
```
purchase = ['No', 'Yes', 'No', 'Yes', 'No', 'Yes', 'No', 'No', 'Yes', 'No']
```

In [78]:
```
df_purchase = pd.DataFrame(purchase, columns=['Purchase'])
```

In [79]:
```
df_purchase
```

Out[79]:

	Purchase
0	No
1	Yes
2	No
3	Yes
4	No
5	Yes
6	No
7	No
8	Yes
9	No

In [80]:
```
df_data1 = pd.DataFrame({'Country' : country, 'Purchase' : purchase})
```

In [81]:
```
df_data1
```

Out[81]:

	Country	Purchase
0	India	No
1	france	Yes
2	Germany	No
3	france	Yes
4	Germany	No
5	India	Yes
6	Germany	No
7	france	No
8	India	Yes
9	Germany	No

In [83]:
```
#Mearging the dataframe
country_data = df_data.merge(df_data1, how='inner', on='Country')
```

In [86]:
```
#saving dataset to csv
country_data.to_csv('country_dataset.csv', index=False)
```

In [87]:
```
country_data
```

Out[87]:

	Country	Age	Salary	Purchase
0	India	38	34000	No
1	India	38	34000	Yes
2	India	38	34000	Yes
3	france	43	45000	Yes
4	france	43	45000	Yes
5	france	43	45000	No
6	Germany	38	54000	No
7	Germany	38	54000	No
8	Germany	38	54000	No
9	Germany	38	54000	No
10	france	34	nan	Yes
11	france	34	nan	Yes
12	france	34	nan	No
13	Germany	54	54500	No
14	Germany	54	54500	No
15	Germany	54	54500	No
16	Germany	54	54500	No
17	India	nan	40000	No
18	India	nan	40000	Yes
19	India	nan	40000	Yes
20	Germany	34	34000	No
21	Germany	34	34000	No
22	Germany	34	34000	No
23	Germany	34	34000	No
24	france	56	23000	Yes
25	france	56	23000	Yes
26	france	56	23000	No
27	India	43	60000	No
28	India	43	60000	Yes
29	India	43	60000	Yes
30	Germany	57	23000	No
31	Germany	57	23000	No
32	Germany	57	23000	No

	Country	Age	Salary	Purchase
33	Germany	57	23000	No

In [88]:
```
df2 = pd.read_csv('country_dataset.csv')
```

In [89]:
```
df2
```

Out[89]:

	Country	Age	Salary	Purchase
0	India	38.0	48000.0	No
1	India	38.0	48000.0	Yes
2	India	38.0	48000.0	No
3	India	35.0	58000.0	No
4	India	35.0	58000.0	Yes
5	India	35.0	58000.0	No
6	India	50.0	88000.0	No
7	India	50.0	88000.0	Yes
8	India	50.0	88000.0	No
9	France	43.0	45000.0	Yes
10	France	43.0	45000.0	No
11	France	43.0	45000.0	Yes
12	France	43.0	45000.0	Yes
13	France	48.0	65000.0	Yes
14	France	48.0	65000.0	No
15	France	48.0	65000.0	Yes
16	France	48.0	65000.0	Yes
17	France	49.0	79000.0	Yes
18	France	49.0	79000.0	No
19	France	49.0	79000.0	Yes
20	France	49.0	79000.0	Yes
21	France	37.0	77000.0	Yes
22	France	37.0	77000.0	No
23	France	37.0	77000.0	Yes
24	France	37.0	77000.0	Yes
25	Germany	30.0	54000.0	No
26	Germany	30.0	54000.0	Yes
27	Germany	30.0	54000.0	No
28	Germany	40.0	NaN	No
29	Germany	40.0	NaN	Yes
30	Germany	40.0	NaN	No
31	Germany	NaN	53000.0	No
32	Germany	NaN	53000.0	Yes

	Country	Age	Salary	Purchase
33	Germany	NaN	53000.0	No

In [90]:
```
#handling missing data
```

In [91]:
```
from sklearn.impute import SimpleImputer
```

In [95]:
```
import numpy as np
```

In [96]:
```
imputer = SimpleImputer(missing_values=np.nan, strategy='mean')
```

In [99]:
```
from sklearn.preprocessing import LabelEncoder
```

In [100…
```
df2.head()
```

Out[100…

	Country	Age	Salary	Purchase
0	India	38.0	48000.0	No
1	India	38.0	48000.0	Yes
2	India	38.0	48000.0	No
3	India	35.0	58000.0	No
4	India	35.0	58000.0	Yes

In [105…
```
encoder = LabelEncoder()
```

In [123…
```
#creating Empty Dataframe
df_trans2 = pd.DataFrame()
```

In [124…
```
df_trans2
```

Out[124… —

In [125…
```
df_trans2['Country'] = encoder.fit_transform(df2['Country'])
```

In [126…
```
df_trans2
```

Out[126…

	Country
0	2
1	2
2	2
3	2
4	2
5	2
6	2
7	2
8	2
9	0
10	0
11	0
12	0
13	0
14	0
15	0
16	0
17	0
18	0
19	0
20	0
21	0
22	0
23	0
24	0
25	1
26	1
27	1
28	1
29	1
30	1
31	1
32	1

	Country
33	1

In [133…

```
df2_trans2['Age'] = df2['Age']
df2_trans2['Salary'] = df2['Salary']
df2_trans2['Purchase'] = encoder.fit_transform(df2['Purchase'])
```

In [134…

```
df_trans2
```

Out[134…

	Country
0	2
1	2
2	2
3	2
4	2
5	2
6	2
7	2
8	2
9	0
10	0
11	0
12	0
13	0
14	0
15	0
16	0
17	0
18	0
19	0
20	0
21	0
22	0
23	0
24	0
25	1
26	1
27	1
28	1
29	1
30	1
31	1
32	1

	Country
33	1

In [135…
```
df_trans2.isnull().sum()
```

Out[135…
```
Country    0
dtype: int64
```

In [137…
```
df2_trans1 = imputer.fit_transform(df2_trans2)
```

In [138…
```
df2_trans1
```

Out[138…
```
array([[2.00000000e+00, 3.80000000e+01, 4.80000000e+04, 0.00000000e+00],
       [2.00000000e+00, 3.80000000e+01, 4.80000000e+04, 1.00000000e+00],
       [2.00000000e+00, 3.80000000e+01, 4.80000000e+04, 0.00000000e+00],
       [2.00000000e+00, 3.50000000e+01, 5.80000000e+04, 0.00000000e+00],
       [2.00000000e+00, 3.50000000e+01, 5.80000000e+04, 1.00000000e+00],
       [2.00000000e+00, 3.50000000e+01, 5.80000000e+04, 0.00000000e+00],
       [2.00000000e+00, 5.00000000e+01, 8.80000000e+04, 0.00000000e+00],
       [2.00000000e+00, 5.00000000e+01, 8.80000000e+04, 1.00000000e+00],
       [2.00000000e+00, 5.00000000e+01, 8.80000000e+04, 0.00000000e+00],
       [0.00000000e+00, 4.30000000e+01, 4.50000000e+04, 1.00000000e+00],
       [0.00000000e+00, 4.30000000e+01, 4.50000000e+04, 0.00000000e+00],
       [0.00000000e+00, 4.30000000e+01, 4.50000000e+04, 1.00000000e+00],
       [0.00000000e+00, 4.30000000e+01, 4.50000000e+04, 1.00000000e+00],
       [0.00000000e+00, 4.80000000e+01, 6.50000000e+04, 1.00000000e+00],
       [0.00000000e+00, 4.80000000e+01, 6.50000000e+04, 0.00000000e+00],
       [0.00000000e+00, 4.80000000e+01, 6.50000000e+04, 1.00000000e+00],
       [0.00000000e+00, 4.80000000e+01, 6.50000000e+04, 1.00000000e+00],
       [0.00000000e+00, 4.90000000e+01, 7.90000000e+04, 1.00000000e+00],
       [0.00000000e+00, 4.90000000e+01, 7.90000000e+04, 0.00000000e+00],
       [0.00000000e+00, 4.90000000e+01, 7.90000000e+04, 1.00000000e+00],
       [0.00000000e+00, 4.90000000e+01, 7.90000000e+04, 1.00000000e+00],
       [0.00000000e+00, 3.70000000e+01, 7.70000000e+04, 1.00000000e+00],
       [0.00000000e+00, 3.70000000e+01, 7.70000000e+04, 0.00000000e+00],
       [0.00000000e+00, 3.70000000e+01, 7.70000000e+04, 1.00000000e+00],
       [0.00000000e+00, 3.70000000e+01, 7.70000000e+04, 1.00000000e+00],
       [1.00000000e+00, 3.00000000e+01, 5.40000000e+04, 0.00000000e+00],
       [1.00000000e+00, 3.00000000e+01, 5.40000000e+04, 1.00000000e+00],
       [1.00000000e+00, 3.00000000e+01, 5.40000000e+04, 0.00000000e+00],
       [1.00000000e+00, 4.00000000e+01, 6.34516129e+04, 0.00000000e+00],
       [1.00000000e+00, 4.00000000e+01, 6.34516129e+04, 1.00000000e+00],
       [1.00000000e+00, 4.00000000e+01, 6.34516129e+04, 0.00000000e+00],
       [1.00000000e+00, 4.15161290e+01, 5.30000000e+04, 0.00000000e+00],
       [1.00000000e+00, 4.15161290e+01, 5.30000000e+04, 1.00000000e+00],
       [1.00000000e+00, 4.15161290e+01, 5.30000000e+04, 0.00000000e+00]])
```

In [139…
```
df2_trans1 = pd.DataFrame(df2_trans1)
```

In [140…
```
df2_trans1.isnull().sum()
```

Out[140…
```
0    0
1    0
2    0
3    0
dtype: int64
```

In [ ]:

```

```

In [ ]:

```

```